Iodine Electric Propulsion for Interplanetary Small Satellite Application

John W. Dankanich[1], Hani Kamhawi[2], James Szabo[3] and Bruce Pote[3]

[1]NASA Marshall Space Flight Center
[2]NASA Glenn Research Center
[3]Busek Co., Inc.

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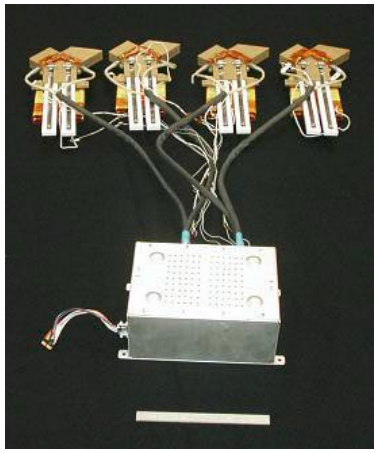
Introduction

- The small satellite market continues to grow
- Propulsion limits the majority of small spacecraft to their deployed orbits
- Science payloads and small satellite subsystems continue to advance
- Propulsion is a key limiter in future missions for small satellites
- High I_{SP} propulsion is demanding for spacecraft resources (power, mass, volume)
- Electric propulsion has been shown to enable a wide range of missions
- Secondary payloads with high ΔV can enable low-cost science missions
- SOA EP has limited applicability to small satellites / secondary payloads
  - High pressure systems required
  - Poor storage density
  - High power demands

Advanced propulsion can enable a new class of missions.

Background

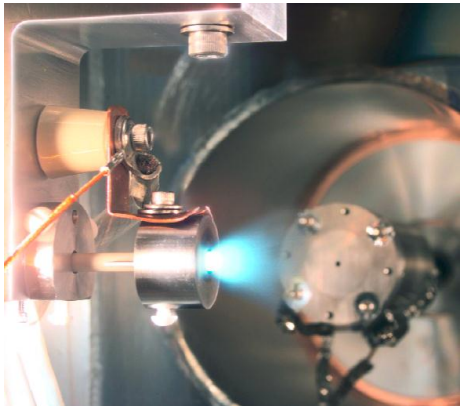
H2O2 - Scharlemann



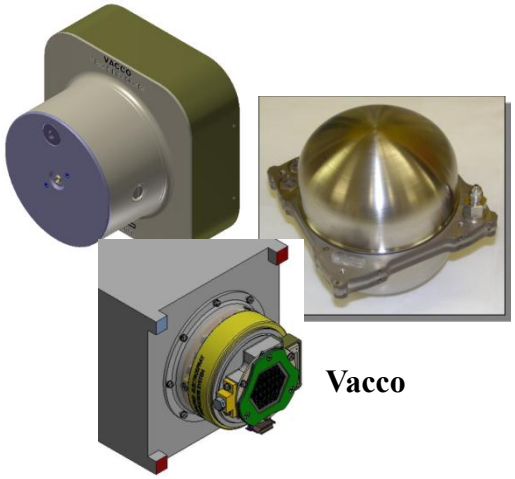
PPT - Aerojet

4G SRM - ATK

MIXI - JPL



VAT – Almeda ASC

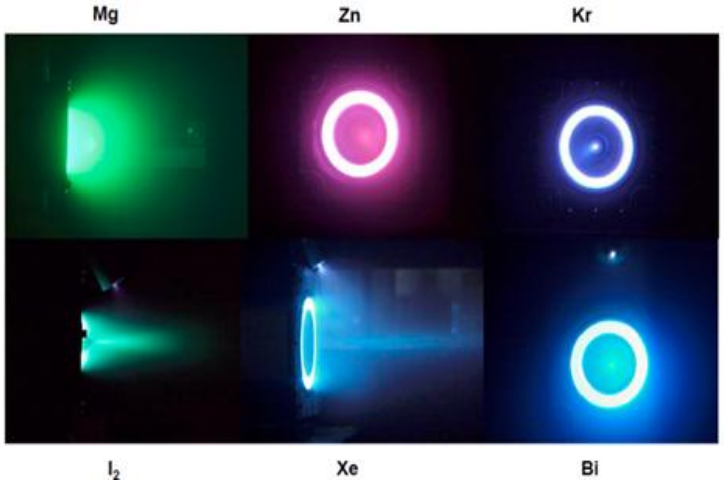


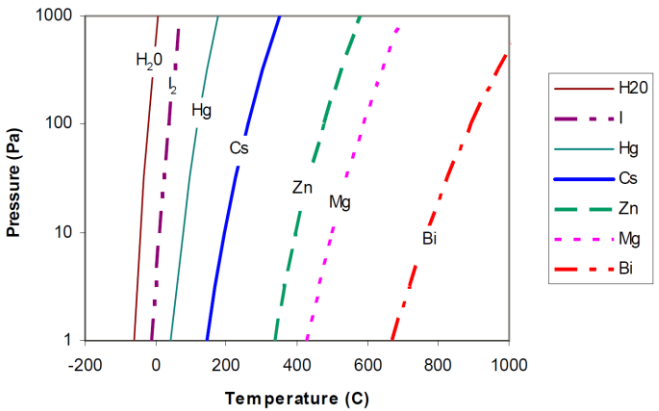
Vacco

A large number of propulsion concepts have been studied and developed to various levels of maturation for small satellite application.

Why Iodine?

- Busek has been leading the exploration of alternative propellants
  - Magnesium
  - Zinc
  - Iodine
  - Bismuth

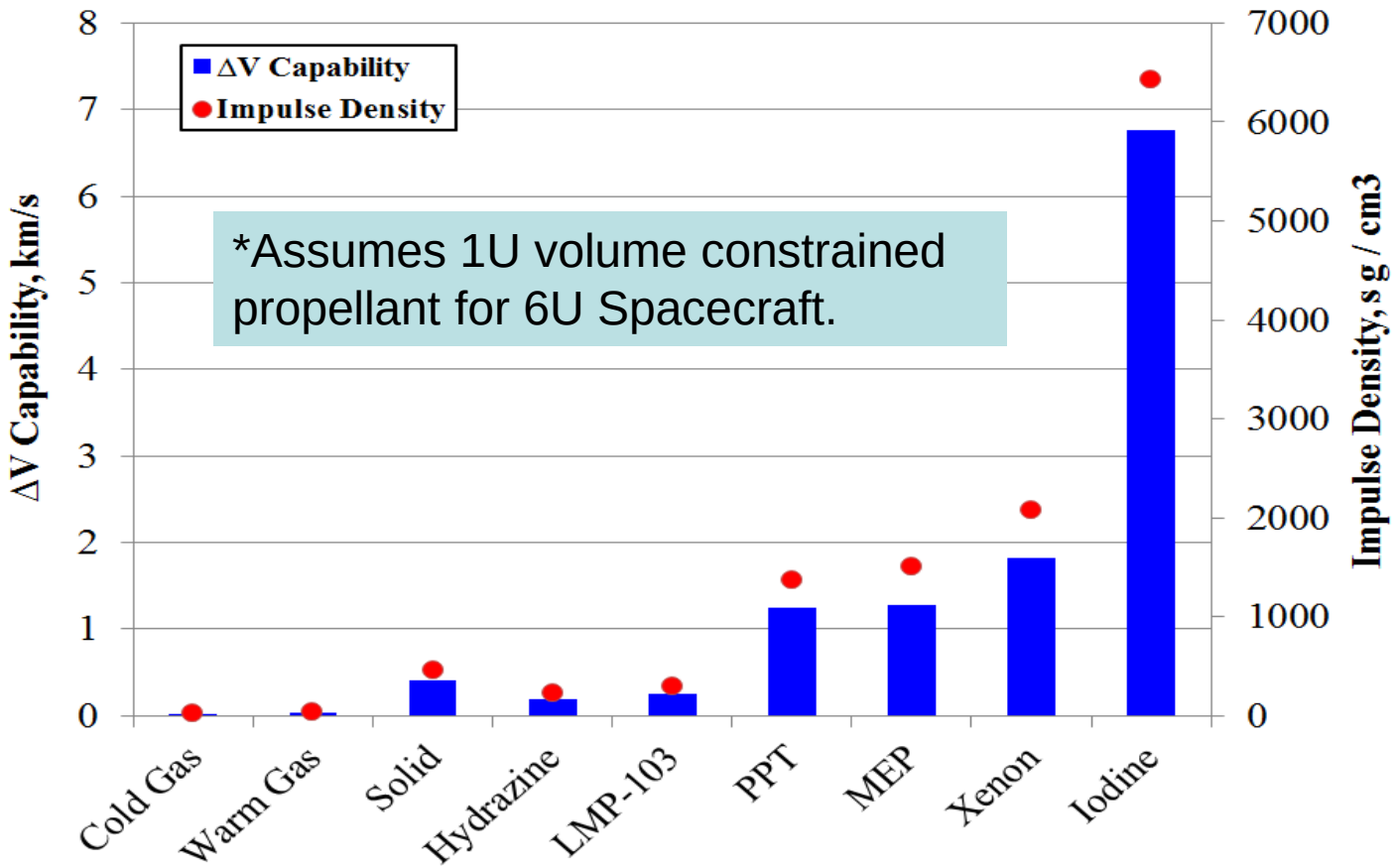




Propellant	Storage Density	Boiling Point, °C	Melting Point, °C	Vapor Pressure @ 20°C
Xe (SOA)	1.6 g/cm^3	-108.1 °C	-111.8 °C	Supercritical (>15MPa)
Iodine	4.9 g/cm^3	184.3 °C	113.7 °C	40 Pa (0.0004 atm)
Bismuth	9.8 g/cm^3	1,564 °C	271.4 °C	Solid
Magnesium	1.74 g/cm^3	1,091 °C	650 °C	Solid

Iodine properties are well suited for volume and power constrained applications.

Why Iodine?

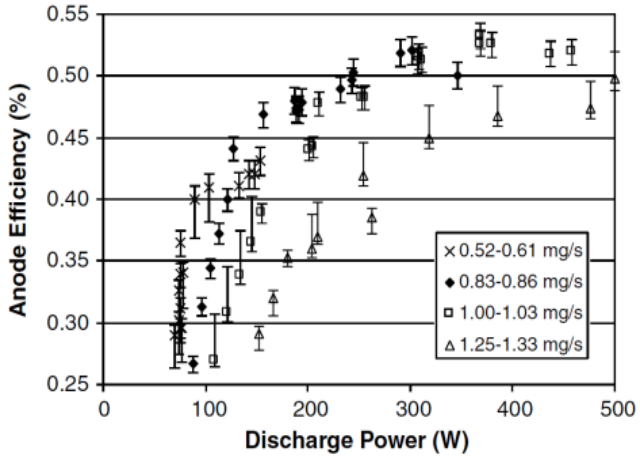


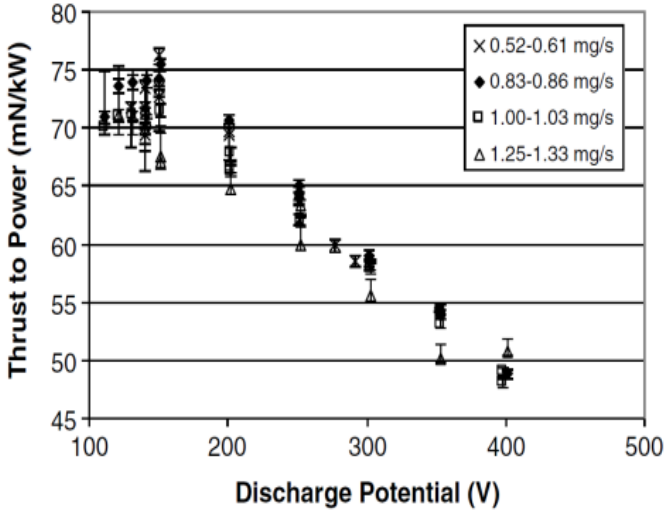
Iodine offers high ΔV capability due to both high specific impulse and high density for volume constrained missions.

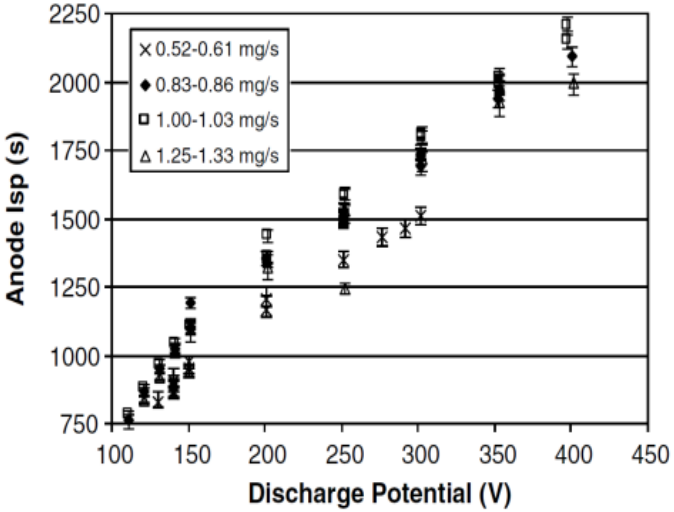
**Figure does not account for mass fraction / supporting subsystems.

Thruster Performance

- Successfully demonstrated BHT-200 w/ Iodine
  - Anode efficiencies of 48%
  - > 70mN/kW
  - > 2000s I_{SP}







An iodine Hall thruster has been demonstrated for near term infusion.

Cathode Development

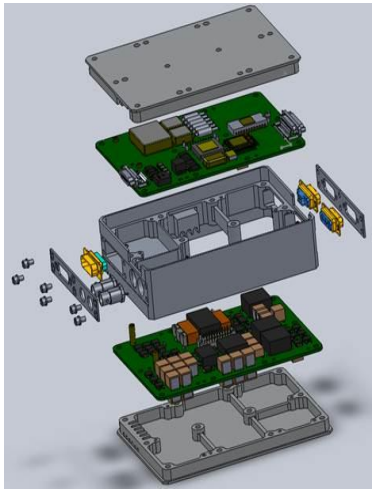
- Cathode short duration testing complete
- NASA Glenn is leading the design and test
  - Risk element for long life demonstration
  - Different emitter materials (LaB6, CeB6, and BaO Tungsten)
  - Long duration testing prior to integrated tests with modified Busek Thruster

Cathode operation with iodine at Busek.

NASA Glenn is performing risk mitigation and long duration testing.

PPU Development

Flight BPU-600



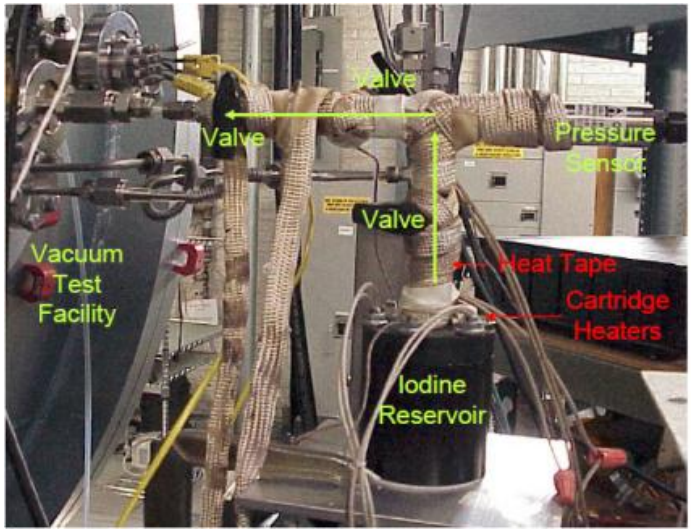
Compact PPU Development

- Iodine Hall thrusters require the same input conditions as xenon thrusters
- Leveraging previous investments
- Optimizing for small satellite application

80% Mass Reduction, 90% Volume Reduction

	Flight Model	Compact PPU
Dimensions	19.6cm x 20.8cm x 20.3 cm	16.5cm x 11.4cm x 4.4 cm
Total Volume	8,300 cm^3	836 cm^3
Electrical Efficiency	94% Peak	95% Peak
Output Voltage	0 - 400V	0 - 400V
Mass	7.23 kg	1.4 kg

Feed System and Tank



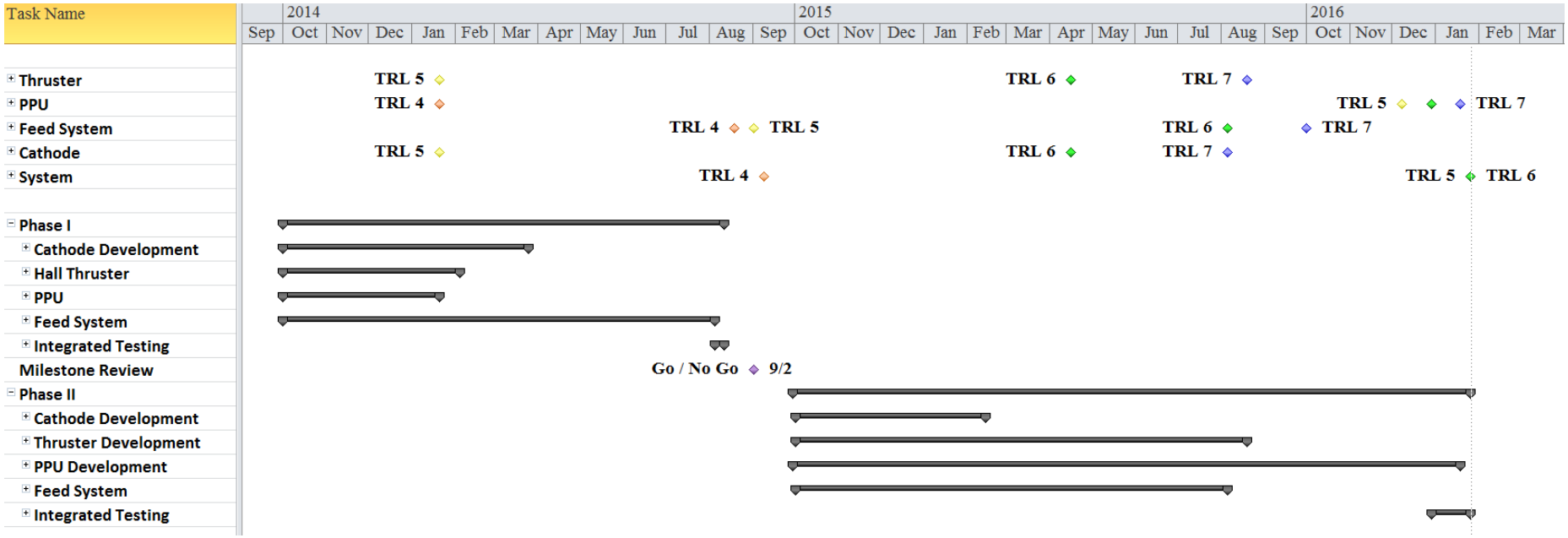
Busek laboratory Iodine feed system

Propellant Tank Comparison		
	Xenon	Iodine
5kg Volume	~3U	~1U
Pressure	> 1000psi	< 15psi
Temperature	30-50^0 C	Ambient
Tank Type	Al/Ti COPV	Sheet Stock
Impact on SC	Large	Small
Cost	$X00k	$X0k

- Ongoing system trades for small satellite feed system optimization
- Tank Pressure controlled vs. Proportional Flow Control
- Valve procurement release
- Evaluating “printed” tanks for conformal shapes to maximize volume

Iodine offers system level advantages and cost saving opportunities

Notional Development Schedule

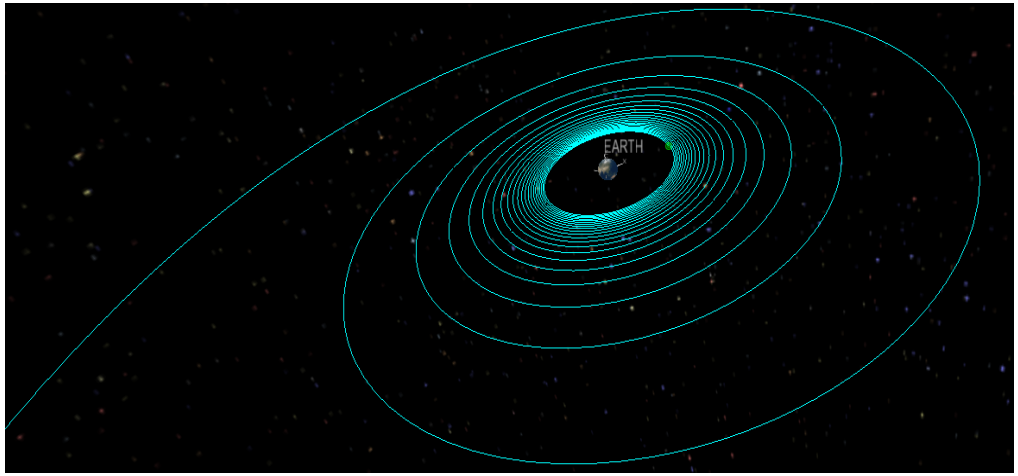


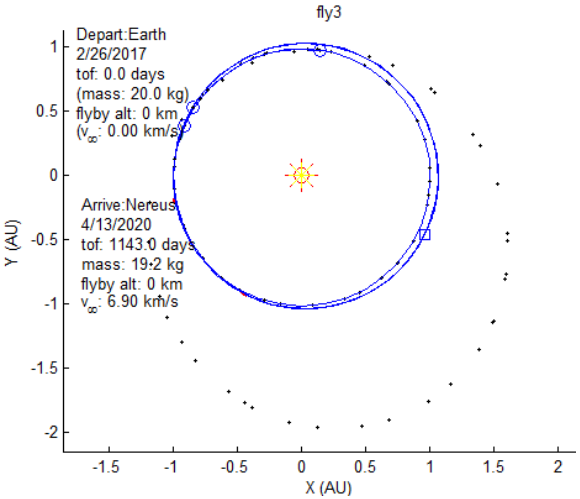
- Leveraging multiple SBIR efforts (NASA and Air Force)
- Leveraging NASA MSFC and Glenn Center Investment Funds

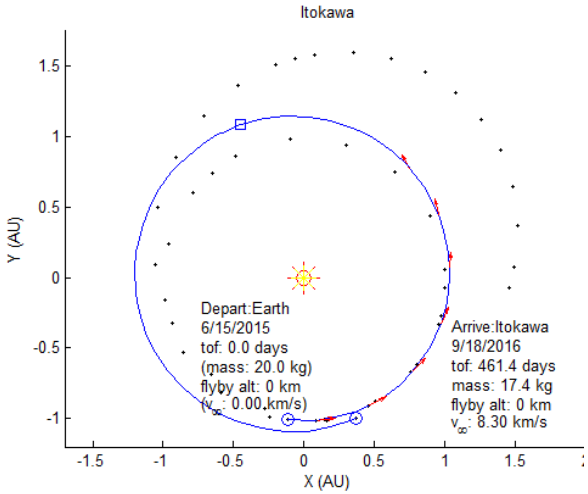
Potential for flight infusion as early as 2016.

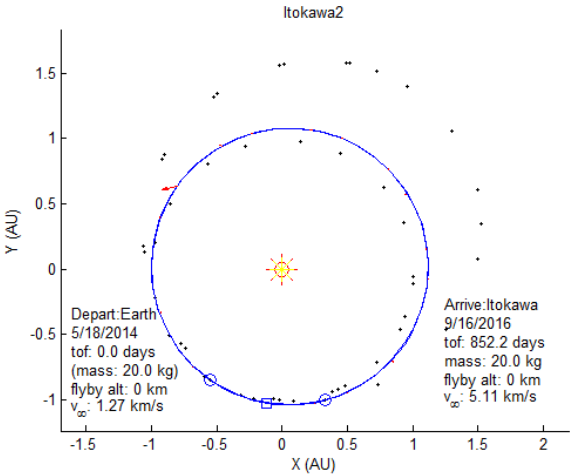
NEA Flybys

- Can escape from GTO or GEO
  - 2kg required from GEO to escape with 6U CubeSat
  - 12U from escape to Nereus w/ 0.8kg of propellant (30W array)
  - Can Leverage lunar assist



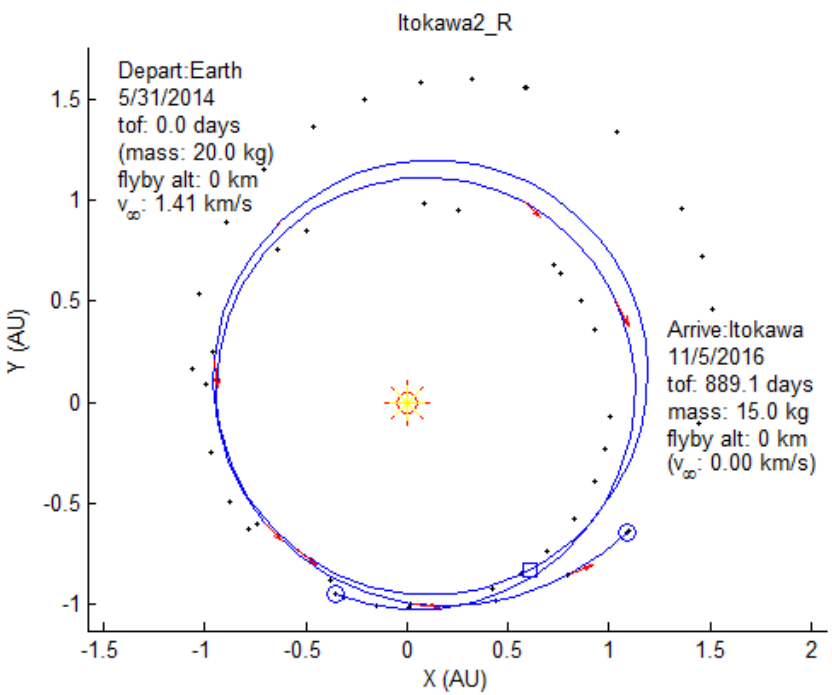


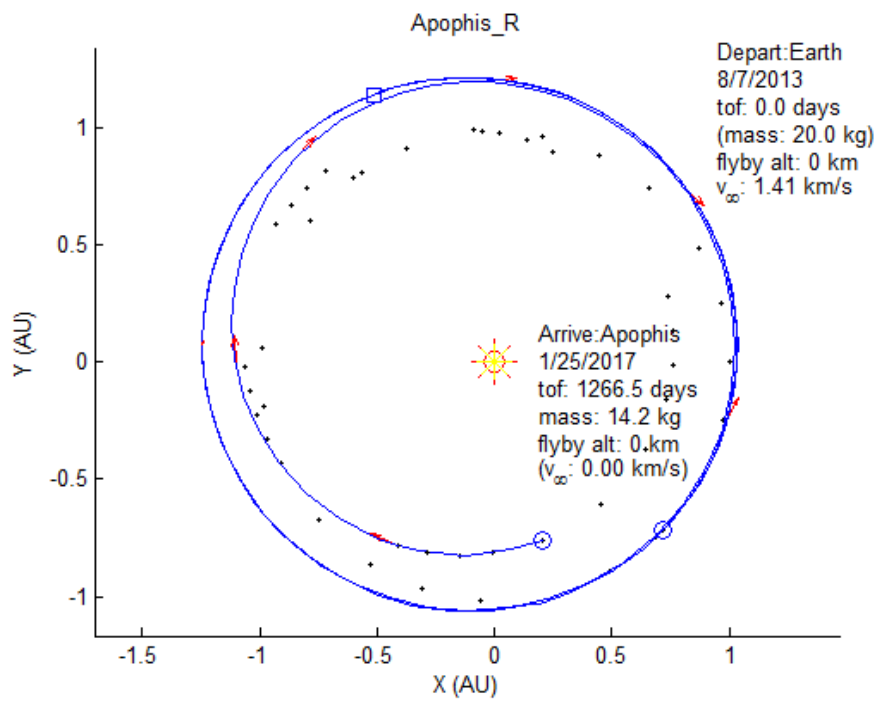




Iodine can enable small sat flyby missions with 6U and larger spacecraft.

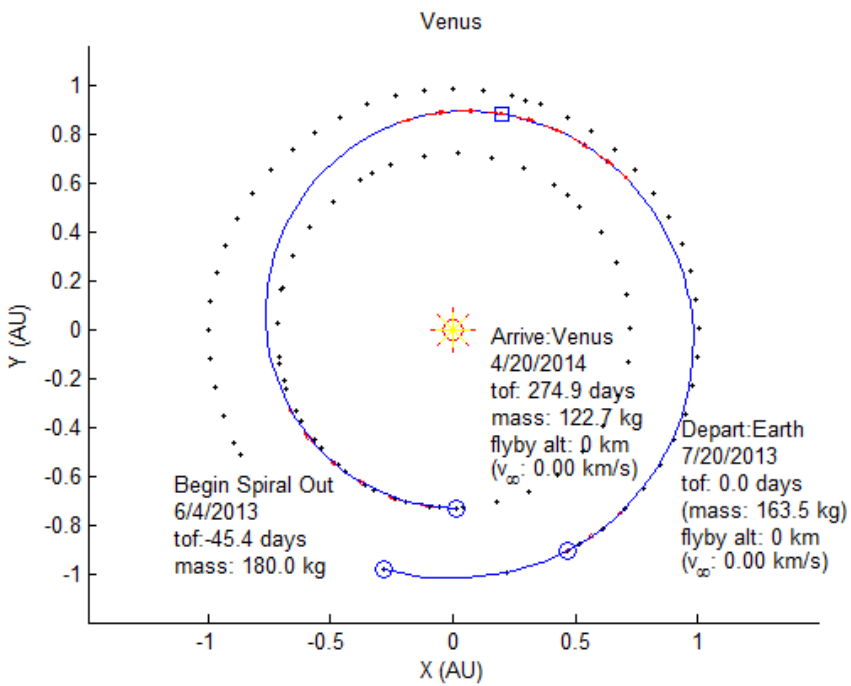
NEA Rendezvous

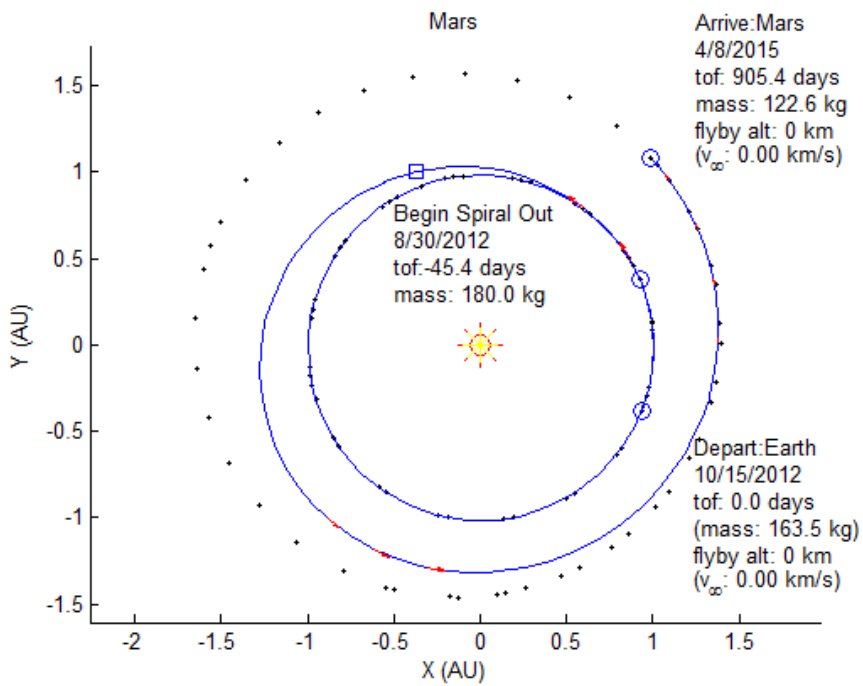




NEA rendezvous missions are enabled by iodine for spacecraft as small as 12U

Venus and Mars from GTO

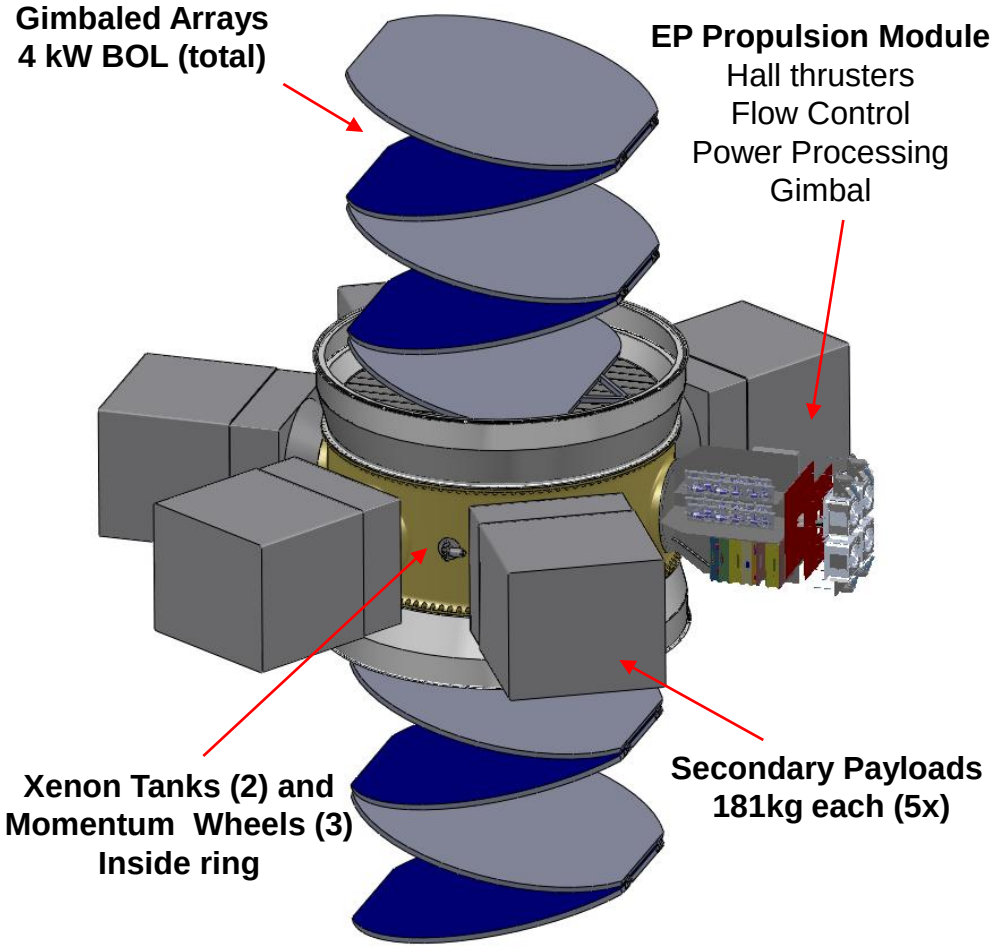


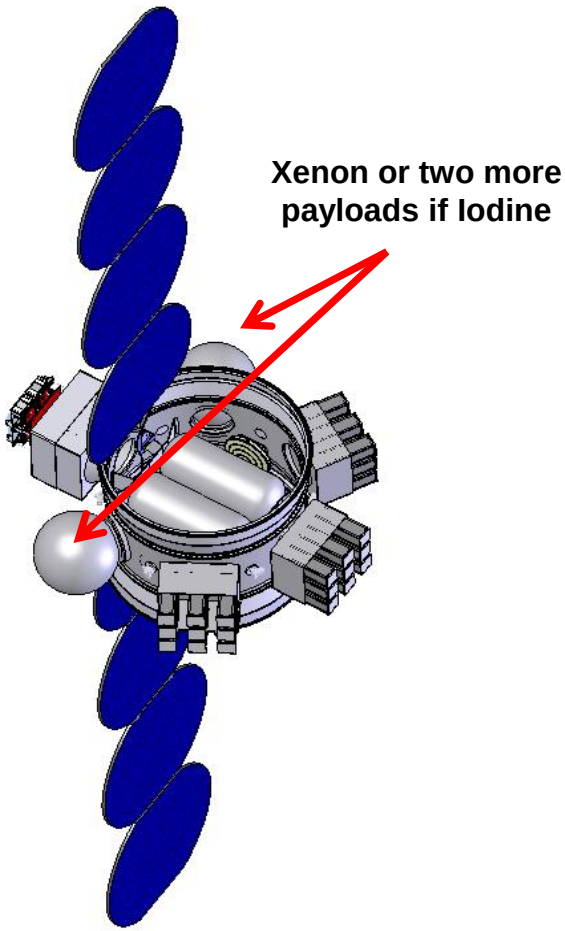


- A single ESPA spacecraft (180kg) can delivery payloads to both Venus and Mars as a secondary GTO payload.

High density iodine can enable Venus and Mars missions as GTO secondary payloads. (Launch costs <$10M)

Transfer Vehicles





- Iodine density can enhance the Xenon ESPA OMS

The ESPA Orbit Maneuvering System can deliver several small sats to strategic orbits of Mars, Venus, asteroids, etc.

Summary

- Iodine Hall thrusters are uniquely advantageous for small satellite applications
  - High density, long duration quiescence, no pressurization required
- Iodine Hall thrusters offer near-term infusion potential
  - Flight heritage to Xenon SOA
  - Leverages flight heritage PPU developments
- Enables high ΔV small satellite missions
  - NEA flyby 6U spacecraft
  - NEA rendezvous 12U spacecraft
  - Small satellite secondary payloads to Venus, Mars, and Asteroids
  - Enables interplanetary secondary payloads with large launch cost savings

Iodine can enable a new class of near-term interplanetary missions.

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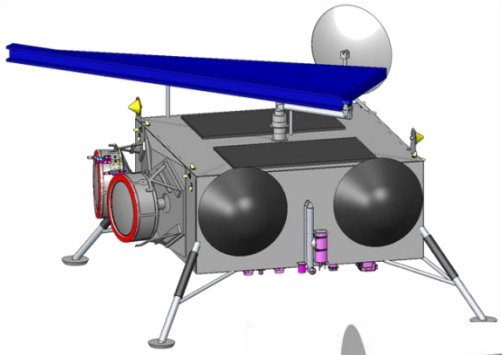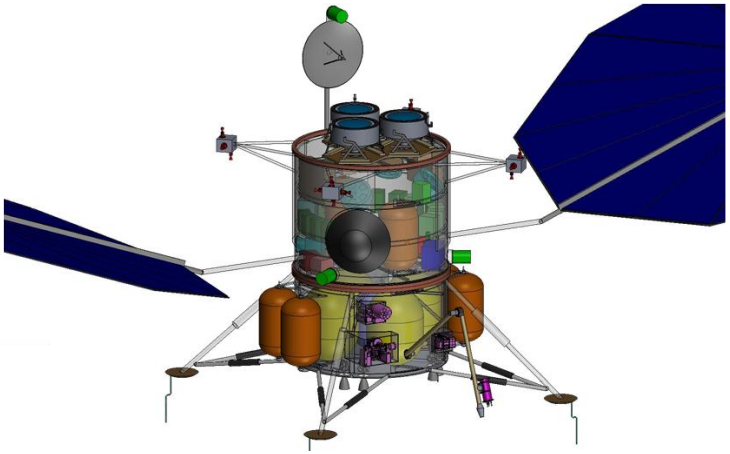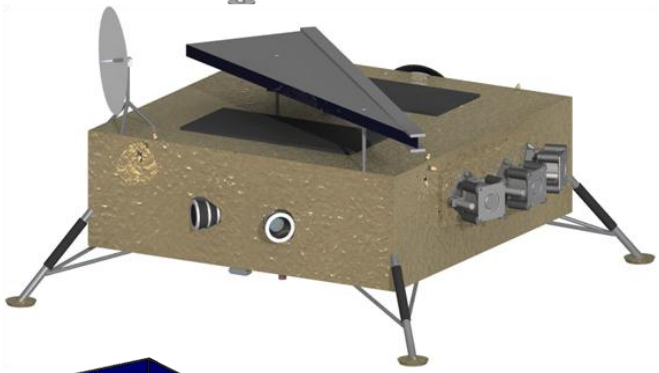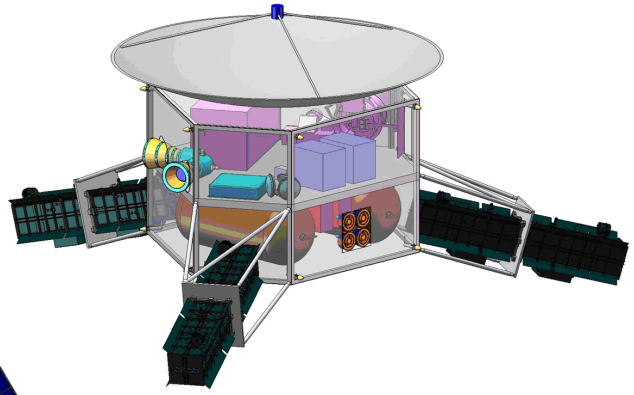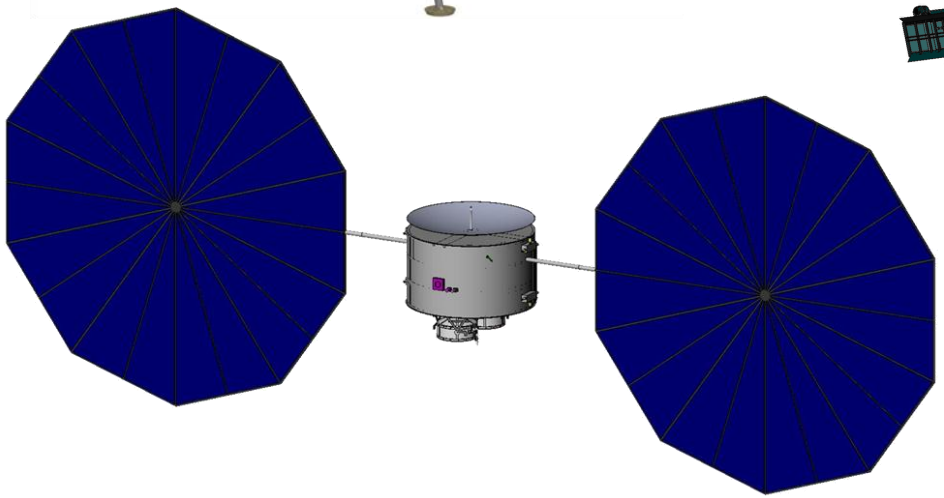
Questions?